

LUT Usage	FF Usage	Latch Usage	WNS
249	193	0	0.965 ns

1. Vivado didn't infer any latches because we were careful to always include a default value in combinational logic, which includes a default branch in case-statements and an else branch for if-statements.
2. We expect the counters to take up a lot of the flip-flops, since there are 27 bit counters in both tick_gen and pulse_gen, a 28 bit accumulator in pulse_gen, and other counters for the required measurements. For LUTs, the most combinational-heavy component is the binary-bcd converter that uses the Double Dabble algorithm, and in the pulse generator, the accumulator increment/wrap logic.